

```
exercise5.py  x
26     string = '\n'.join(lines)
27     self.words = string
28
29     def swap_words(self):
30         self.string_to_list()
31         for line in self.words:
32             for i in range(len(line)):
33                 if (i + 1) % 5 == 0:
34                     word = line[i]
35                     line[i] = 'HELLO'
36         self.list_to_string()
37
38     class Stars(Substitute):
39     def swap_words(self):
40         self.string_to_list()
41         for line in self.words:
42             for i in range(len(line)):
43                 if (i + 1) % 3 == 0:
44                     word = line[i]
45                     line[i] = '*'*len(word)
46         self.list_to_string()
47
48 #s = Substitute(source_file, answer_file)
49 s = Stars(source_file, answer_file)
50 s.swap_words()
51 print(s.words)
```

Polymorphism Exercise 5

Exercise 5

The `Substitute` class reads a text file and replaces every fifth word with the string `HELLO`.

▼ Parent class output

To see the output from the parent class, add the following code to your script and run it.

```
s = Substitute(source_file, answer_file)
s.swap_words()
print(s.words)
```

Create the class `Stars` which is a child class of `Substitute`. Then override the `swap_words` method so that every third word is replaced by a series of `*`. If the word has three letters then it should be replaced with `***`. The number of `*` should match the number of characters in the word. Write the new string to `self.answer_file`.

Important

Keep the following things in mind as you write your code. Changes should only be made to the `Stars` class; do not alter the code for the file variables and the `Substitute` class.

- The `source_file` attribute represents the file you will read.
- The `answer_file` attribute represents the file to which you will write your new text.
- The `string_to_list` method converts the string (the text file) into a 2D list. Each inner list represents a sentence. The elements of each inner list are the words that make up that sentence.
- The `list_to_string` method converts the 2D list back into a string.
- When you override the `swap_words` methods, be sure to start with `string_to_list` and use `list_to_string` before writing to the file.
- The text used for this activity are the first four sentences of Jane Austen's *Pride and Prejudice*.

TRY IT

```
It is * truth universally ***** that a *****
man in ***** of a *** fortune, must ** in want
** a wife.
However little ***** the feelings ** views of **** a
man *** be on *** first entering * neighbourhood,
this ***** is so ***** fixed in *** minds of ***
surrounding families, **** he is ***** the
rightful ***** of some *** or other ** their
daughters.
“My dear *** Bennet,” said *** lady to *** one day,
***** you heard **** Netherfield Park ** let at
*****
Mr. Bennet ***** that he *** not.
```

See your answer

▼ Expected output

The `answer_file` should look like the text below.

```
It is * truth universally ***** that a ***** man in ***** of a **** f
However little ***** the feelings ** views of **** a man *** be on *** first enteri
“My dear *** Bennet,” said *** lady to *** one day, ***** you heard **** Netherfiel
Mr. Bennet ***** that he *** not.
```

Submit your code for evaluation

Here is one possible solution:

```
source_file = '/home/codio/workspace/code/polymorphism/text_1_exercise5.
answer_file = '/home/codio/workspace/code/polymorphism/answer_exercise5.

class Substitute:
    def __init__(self, source_file, answer_file):
        self.source_file = source_file
        self.answer_file = answer_file
        self.words = None

    def string_to_list(self):
        '''Read text file, turn it into a
        2D list of words for each line'''
        words = []
        with open(self.source_file, 'r') as file_object:
            lines = file_object.read().split('\n')
            for line in lines:
                words.append(line.split())
            self.words = words

    def list_to_string(self):
        '''Convert 2D list back into a
        string with newline characters'''
        lines = []
        for line in self.words:
            lines.append(' '.join(line))
        string = '\n'.join(lines)
        self.words = string

    def swap_words(self):
        self.string_to_list()
        for line in self.words:
            for i in range(len(line)):
                if (i + 1) % 5 == 0:
                    word = line[i]
                    line[i] = 'HELLO'
        self.list_to_string()

class Stars(Substitute):
    def swap_words(self):
        self.string_to_list()
        for line in self.words:
            for i in range(len(line)):
                if (i + 1) % 3 == 0:
                    word = line[i]
                    line[i] = '*' * len(word)
        self.list_to_string()
        file = open(self.answer_file, 'w')
        file.writelines(self.words)
        file.close()

s = Stars(source_file, answer_file)
s.swap_words()
```

Check It!

LAST RUN on 12/28/2023, 2:58:01 AM

Test passed

Mark as Completed

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